

MOHAMED AYMAN MOHAMED

Backend Software Engineer

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[GitHub: MohamedAyman23](#) | [LinkedIn: mohamed-ayman](#) | Military Service: Completed

PROFILE

Backend Software Engineer specializing in Java and Spring Boot, with hands-on experience designing and building RESTful APIs, database-driven systems, authentication, caching, asynchronous messaging, document generation, and production deployments. Independently built and deployed the backend of a live B2B agricultural export platform, while currently developing an Electronic Document Management System (EDMS) and the Raabit platform.

TECHNICAL SKILLS

Languages & Core CS: Java, C, SQL, Data Structures & Algorithms, Collections Framework, OOP

Backend: Spring Core, Spring MVC, Spring Boot 4.1.0, Spring Data JPA, Spring Security

Databases & Persistence: PostgreSQL, MySQL, SQL Server, Flyway

Caching & Messaging: Redis, RabbitMQ

Storage & Deployment: Cloudflare R2 (S3-compatible), Docker, Docker Compose, Linux server administration

Document Generation: Thymeleaf, openhtmltopdf, Apache POI

API & Tools: RESTful APIs, Swagger/OpenAPI, Postman, multipart file uploads

Security: JWT, RBAC, token blacklisting, BCrypt

Testing & Collaboration: JMeter, Manual/API Testing, Git/GitHub, Jira/Trello

PROFESSIONAL EXPERIENCE

Almanara Agricultural Crops | Backend Developer | July 2025 – Present

Ismailia, Egypt | Part-time, concurrent with SBS role

- Independently built and deployed the backend of Almanara Export, a live B2B agricultural export platform, from architecture design through production deployment.
- Architected and deployed a multi-container Spring Boot environment using Docker Compose with PostgreSQL, Redis, RabbitMQ, and Cloudflare R2; owned server provisioning and production configuration.
- Currently developing the Almanara EDMS, including dynamic data selection, PDF/Excel generation, document storage, versioning, and secure file delivery.
- Implemented dynamic data retrieval using JPA Criteria API and Spring Data JPA Specifications, supporting flexible filtering, pagination, and user-selected tables, rows, and fields.
- Implemented asynchronous email delivery with RabbitMQ and Microsoft Graph API, including retry handling and dead-letter queue (DLQ) processing.
- Implemented JWT authentication, RBAC, refresh tokens, and Redis-based token blacklisting with TTL for stateless authentication and secure logout.
- Designed PostgreSQL schemas and Flyway migrations, JSONB-based flexible document selections, GIN indexes, atomic numbering with optimistic locking, and reset policies.
- Built 40+ RESTful APIs across platform modules and owned deployment and ongoing maintenance of the live system.

Specialized Business Solutions (SBS) | Software Engineer | June 2025 – Present

Smart Village, Egypt | Part-time, concurrent with Almanara role

- Performed manual, API (Postman/Swagger), and performance (JMeter) testing across Government, Banking, HR, and E-Commerce systems spanning multiple client projects.
- Automated chatbot response validation with Python, reducing manual regression-testing effort for conversational flows.

PROJECTS

Export Documentation Management System (EDMS) | Under Development

A specialized Export Documentation Management System currently under development, allowing users to select combinations of tables, rows, and columns and generate PDF/Excel documents without requiring schema changes for every document-selection combination.

- Dynamic data fetching and filtering using JPA Criteria API and Spring Data JPA Specifications.
- PDF and Excel generation using Thymeleaf, openhtmltopdf, and Apache POI.
- Cloudflare R2 object storage with secure pre-signed URLs and document versioning.
- Asynchronous email processing with RabbitMQ and Microsoft Graph API, including retry and DLQ handling.
- JWT authentication, RBAC, Redis token blacklist with TTL, and refresh-token support.
- Atomic document numbering using optimistic locking with configurable yearly/monthly reset policies.
- JSONB-based storage for flexible document selections without requiring schema changes.

Tech Stack: Java 21, Spring Boot 4.1.0, Spring Data JPA, PostgreSQL, Redis, RabbitMQ, Cloudflare R2, Docker, Flyway, Thymeleaf, openhtmltopdf, Apache POI

Raabit Platform | Backend System | Under Development

A B2B platform currently under development for connecting agricultural exporters with business partners and supporting digital workflows around the export ecosystem.

- Backend architecture and core services are being designed and implemented with scalability and maintainability in mind.
- Designed as a separate platform alongside the EDMS initiative, with the goal of supporting exporter-facing business workflows.

Status: Under active development — technologies and feature set are evolving.

EDUCATION

B.Sc. in Computer Engineering – Suez Canal University | Sep 2018 – Jul 2023

Major: Computer and Control Systems | GPA: Very Good

- Graduation Project: Student Smart Tracking System (IoT/Embedded) – Team Leader. Engineered embedded hardware and a mobile app to track attendance via RFID. Grade: Excellent.

CERTIFICATIONS & PROFESSIONAL TRAINING

- Spring, Spring Boot, and Spring Security – Udemy (EasyBytes)
- ISTQB Foundation Level for Agile Testers – February 2025
- Test Automation with Selenium WebDriver (Java) – December 2025 – Present
- ISTQB Foundation Level – December 2024
- API Testing with Postman – Eng. Tarek Roshdy, March 2025

LANGUAGES

Arabic – Native | English – Intermediate (Professional Working Proficiency)